

Congenital Myasthenic Syndrome 7 (CMS7 / SYT2-Related Presynaptic Myasthenia): A Comprehensive Disease Characterization

MONDO: MONDO:0014468 · **OMIM:** #616040 (CMS7A, dominant) / #619461 (CMS7B, recessive) ·
Gene: SYT2 (synaptotagmin-2) · **Category:** Mendelian, rare

Summary

Congenital Myasthenic Syndrome 7 (CMS7) is an ultra-rare, genetically determined **presynaptic** disorder of neuromuscular transmission caused by germline variants in **SYT2 (synaptotagmin-2)**, the protein that serves as the calcium sensor for fast, synchronous acetylcholine (ACh) release at the neuromuscular junction (NMJ). Because the primary lesion sits in the nerve terminal rather than the muscle endplate, CMS7 behaves clinically and electrophysiologically like a genetic mimic of the autoimmune Lambert–Eaton myasthenic syndrome (LEMS): low compound muscle action potential (CMAP) amplitudes at rest that facilitate (increment) after brief exercise or high-frequency stimulation, together with a characteristically prolonged post-tetanic potentiation. The disease is important out of proportion to its rarity because it is **treatable** — patients respond to the potassium-channel blocker 3,4-diaminopyridine (3,4-DAP) and to acetylcholinesterase inhibitors such as pyridostigmine.

CMS7 has **two inheritance modes with distinct severity**. Autosomal **dominant** disease arises from heterozygous missense or in-frame variants clustered in the **C2B calcium-binding domain** (canonically p.Asp307Ala and p.Pro308Leu), which act by a **dominant-negative** mechanism; these patients present with slowly progressive or non-progressive distal lower-limb weakness and wasting, foot deformities (pes cavus), fatigable weakness, hyporeflexia/areflexia with post-exercise reflex potentiation, and a phenotype that can mimic a distal hereditary motor neuropathy/Charcot–Marie–Tooth disease. Autosomal **recessive** disease

arises from **biallelic loss-of-function** variants (frameshift, nonsense, splice-site, large deletion), frequently in consanguineous families, and is substantially more severe — congenital-onset hypotonia, profound weakness, areflexia, bulbar dysfunction, and variable respiratory involvement.

This report synthesizes 13 confirmed findings drawn from 29 papers, including the seminal disease-gene reports (Herrmann 2014, Whittaker 2015), recessive-form descriptions (2019–2021), a *Drosophila* causality model, comprehensive CMS reviews (35- and 40-gene reviews), UK epidemiology, and gnomAD constraint analysis. Across the sections below, CMS7 is characterized at the level of disease identity, etiology, phenotype, molecular genetics, pathophysiology (presented as an explicit causal chain), affected anatomy, temporal course, inheritance/population genetics, diagnostics, prognosis, treatment, prevention, and model organisms. Where information is genuinely unavailable for this ultra-rare disease (e.g., formal quality-of-life instrument data, natural-history registries specific to SYT2), this is stated explicitly.

1. Disease Information

What it is. CMS7 is a Mendelian congenital myasthenic syndrome in which fast synchronous neurotransmitter release at the NMJ fails because of defects in the presynaptic calcium sensor synaptotagmin-2. It is one of the small group of **presynaptic** CMS subtypes that phenocopy LEMS (the "CMS-LEMS" group), alongside AGRN, MUNC13-1/UNC13A, VAMP1, and LAMA5 (F004; [PMID: 29696584](#)).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0014468
OMIM	#616040 (CMS7A, autosomal dominant); #619461 (CMS7B, autosomal recessive)
Gene (HGNC)	SYT2, HGNC:11510
NCBI Gene	127833
Ensembl	ENSG00000143858
UniProt	Q8N9I0
Cytoband	1q32.1

Synonyms / alternative names. CMS7; SYT2-related disease; SYT2-CMS; presynaptic congenital myasthenic syndrome 7; autosomal-dominant Lambert-Eaton-like myasthenic syndrome with nonprogressive motor neuropathy (dominant form). Gene aliases: **CMS7**, **CMS7A**, **CMS7B**, **MYSPC**, **SytII** (F008).

Data provenance. Essentially all knowledge derives from **aggregated disease-level resources** and **individual published case reports/small case series** (~10–15 families worldwide), not from EHR/population phenotyping — a direct consequence of the disease's extreme rarity.

2. Etiology

Primary cause — genetic. CMS7 is a monogenic disorder caused entirely by pathogenic germline variants in **SYT2**. There is no environmental, infectious, or acquired etiology; it is not autoimmune (distinguishing it mechanistically from myasthenia gravis and from paraneoplastic LEMS). Two allelic mechanisms operate (F001, F010):

- **Dominant** heterozygous **C2B-domain missense/in-frame** variants → dominant-negative poisoning of release.
- **Recessive** biallelic **loss-of-function** variants → absence of functional SYT2.

Genetic risk factors. The causal variants themselves are the risk factor. For the recessive form, **consanguinity** is a major contributing factor (multiple reported homozygous cases in consanguineous families; F010; [PMID: 32250532](#), [PMID: 36722210](#)). No susceptibility loci, GWAS signals, or modifier genes have been established for this Mendelian disease.

Environmental / lifestyle / protective factors. None established. As a channel/release Mendelian disorder, there are no known dietary, occupational, toxic, or lifestyle risk or protective factors, and no gene–environment interactions have been reported. (Not applicable / not available.)

Infectious agents. Not applicable — CMS7 is non-infectious. Intercurrent respiratory infections can precipitate crises in the severe recessive form (a trigger of morbidity, not a cause).

3. Phenotypes

CMS7 phenotypes are **clinical signs and physical manifestations** of impaired neuromuscular transmission plus a secondary neurogenic/motor-neuropathy component. Severity is bimodal by inheritance mode (F005).

Phenotype	Type	Onset	Severity/ Course	Frequency	Suggested HPO
Fatigable muscle weakness	Clinical sign	Congenital–childhood (recessive) to childhood/adult (dominant)	Fluctuating, fatigable	Core / typical	HP:0003473 (Fatigable weakness)
Distal lower-limb weakness & wasting	Physical	Childhood (dominant)	Slowly progressive/stable	Typical in dominant	HP:0002460 (Distal muscle weakness); HP:0003693 (Distal amyotrophy)
Foot deformities / pes cavus	Physical	Childhood	Stable structural	Frequent in dominant	HP:0001761 (Pes cavus)
Neonatal hypotonia	Clinical sign	Neonatal	Severe in recessive	Frequent in recessive	HP:0001319 (Neonatal hypotonia)
Hyporeflexia / areflexia	Clinical sign	Variable	With post-exercise reflex potentiation	Typical	HP:0001315 (Reduced tendon reflexes); HP:0001284 (Areflexia)
Ptosis / ocular weakness	Clinical sign	Variable	Fatigable	Variable	HP:0000508 (Ptosis); HP:0000602 (Ophthalmoplegia)
Bulbar dysfunction (weak cry, dysphonia, dysphagia)	Clinical sign	Neonatal/infancy (recessive)	Moderate–severe	Recessive-predominant	HP:0002019 (Feeding difficulties)
Respiratory insufficiency /	Clinical sign		Severe; morbidity/		HP:0002093 (Respiratory

Phenotype	Type	Onset	Severity/ Course	Frequency	Suggested HPO
recurrent infections		Infancy (recessive)	mortality driver	Recessive-predominant	insufficiency); HP:0002205 (Recurrent respiratory infections)
Delayed motor development	Clinical sign	Infancy	Variable	Recessive-predominant	HP:0001270 (Motor delay)
Elevated CK (mild)	Lab abnormality	—	e.g., CPK 501 U/L	Occasional	HP:0003236 (Elevated creatine kinase)

Key supporting evidence (F005): a 2025 case-based review synthesizing 28 cases describes SYT2-related disease as *"characterized by distal muscle atrophy in the lower limbs, foot deformities and, in some cases, neonatal hypotonia,"* with most mutations in the C2B domain (PMID: 41331967). The dominant-vs-recessive severity gradient is explicit: *"The recessive form of CMS caused by a SYT2 mutation showed far more severe clinical manifestations than the dominant form"* (PMID: 34037996). Distal weakness with pes cavus can mimic distal hereditary motor neuropathy/CMT (PMID: 33320396).

Quality-of-life impact. No disease-specific EQ-5D/SF-36/PROMIS data exist for CMS7 (not available). Qualitatively, the dominant form causes gait impairment and fatigue affecting mobility and daily function; the recessive form causes major disability with respiratory and feeding needs. A salbutamol trial in NMJ-involving motor neuropathies (including SYT2) recorded patient-reported **fatigue** as a meaningful endpoint (F007; PMID: 36869887).

4. Genetic / Molecular Information

Causal gene. SYT2 (synaptotagmin-2), HGNC:11510, NCBI Gene 127833, Ensembl ENSG00000143858, UniProt Q8N9I0, chromosome **1q32.1** (GRCh38 chr1:202,590,596–202,710,526, minus strand) (F008).

Protein architecture. Synaptotagmin-2 is a synaptic-vesicle transmembrane protein with an N-terminal intravesicular/luminal region, a single transmembrane domain, and **two cytoplasmic tandem C2 domains (C2A and C2B)** that bind Ca^{2+} and phospholipids/SNAREs (F008). SYT2 is the major synaptotagmin isoform at the human NMJ and one of only three (SYT1, SYT2, SYT9) that serve as Ca^{2+} sensors for **fast synchronous** release (F002; [PMID: 17521570](#)).

Pathogenic variants and their mechanisms.

Variant (protein / cDNA)	Domain	Type	Inheritance	Mechanism	Reference
p.Asp307Ala (c.920A>C)	C2B	Missense, disrupts essential Ca^{2+} -binding Asp	AD	Dominant-negative	PMID: 25192047
p.Pro308Leu (c.923C>T)	C2B	Missense	AD	Dominant-negative	PMID: 25192047
p.361_365del (c.1082_1096del)	C2B	In-frame deletion, de novo	AD	Dominant-negative	PMID: 33320396
p.Arg397Serfs*37 (c.1191delG)	C2B C-terminus	Frameshift	AR (homozygous)	Loss of function	PMID: 32250532
c.465+1G>A	splice donor	Splice	AR (homozygous)	Loss of function	PMID: 33659639
c.328_331dup	—	Frameshift	AR (homozygous)	Loss of function	PMID: 33659639
Large deletion exons 2–9	multi-exon	Structural	AR (homozygous)	Loss of function	PMID: 36722210

Variant classification & population frequency. Reported pathogenic variants are **private/ultra-rare** and effectively **absent or singleton** in population databases, consistent with high penetrance and de novo/consanguineous origins (F013). gnomAD constraint for SYT2 (F013): **pLI = 0.70**, **LOEUF (upper) = 0.58**, **LoF-z = 3.34** (intolerant to complete LoF), **missense-z = 2.62**

(missense-constrained), synonymous- $z = 0.98$ (neutral, as expected). This constraint profile explains why dominant missense variants are pathogenic and why biallelic LoF produces severe disease.

Somatic vs germline. All variants are **germline**. No somatic disease role.

Functional consequences. Dominant C2B missense/in-frame variants = **dominant-negative**; recessive truncating/splice/structural variants = **loss of function** (F008, F010; [PMID: 34037996](#), [PMID: 32776697](#)).

Modifier genes / epigenetics / chromosomal abnormalities. No modifier genes, epigenetic marks, or large chromosomal abnormalities (beyond the single-gene exonic deletion above) are established for CMS7 (not available). One mechanistic aside: in cortical neurons, calmodulin transcriptionally suppresses *SYT2*, an inverse rostral-caudal expression relationship — relevant to *SYT2* biology but not established as a disease modifier ([PMID: 20729199](#)).

5. Environmental Information

Environmental, lifestyle, and infectious factors. Not applicable to disease causation — CMS7 is purely Mendelian. There are no reported toxin, radiation, occupational, dietary, smoking, alcohol, or exercise contributions to disease risk. Intercurrent **respiratory infections** are clinically relevant only as **precipitants of crisis/morbidity** in the severe recessive form, not as etiologic agents (F012; [PMID: 41331967](#)).

6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

Dominant (C2B missense/in-frame) branch: 1. Heterozygous **C2B-domain missense/in-frame variant** in *SYT2* (e.g., p.Asp307Ala disrupting an essential Ca^{2+} -coordinating aspartate) **produces** a mutant synaptotagmin-2 protein that still incorporates into the release machinery. 2. The mutant protein **acts dominant-negatively**, poisoning the Ca^{2+} -sensing/SNARE-coupling function of the release apparatus even in the presence of one wild-type allele (demonstrated in *Drosophila*, where co-expression with one wild-type copy reproduces disease) → **impaired**

coupling of Ca^{2+} influx to synchronous synaptic-vesicle fusion. 3. This **results in** reduced quantal content (fewer ACh quanta released per nerve impulse), most evident at low stimulation frequencies. 4. Reduced ACh release **leads to** an endplate potential that intermittently fails to reach threshold → **intermittent neuromuscular transmission failure** → fatigable weakness. 5. Because residual Ca^{2+} accumulates in the terminal during repetitive/high-frequency activity, release transiently improves → **facilitation/post-exercise increment and prolonged post-tetanic potentiation** (the LEMS-like signature). 6. Chronic presynaptic dysfunction **is associated with** a secondary distal **motor neuropathy/terminal remodeling** → distal wasting, pes cavus, areflexia → the clinical phenotype.

Recessive (biallelic LoF) branch: 1. **Biallelic loss-of-function variants** (frameshift/nonsense/splice/deletion) **result in** absent or non-functional SYT2 protein. 2. Loss of the principal fast Ca^{2+} sensor **leads to** profound impairment of synchronous ACh release (more severe presynaptic failure than the dominant form). 3. This **results in** congenital-onset severe weakness, hypotonia, bulbar and respiratory compromise. (*The C-terminal recessive variant p.Arg397Serfs37* disrupts a region crucial for synaptotagmin–SNARE interaction and exocytosis — inferred from structural modeling on the rat Syt1 C2B template.)*

Molecular / cellular detail (checklist)

- **Molecular pathways / biochemical defect.** Ca^{2+} -triggered synaptic-vesicle exocytosis; SYT2 is the Ca^{2+} sensor that couples voltage-gated Ca^{2+} influx to SNARE-mediated membrane fusion. The core defect is an **ion-sensor/receptor-coupling failure**, not an enzyme deficiency (F002; [PMID: 17521570](#)). Suggested GO: **GO:0016079** (synaptic vesicle exocytosis), **GO:0017156** (calcium-ion-regulated exocytosis), **GO:0005544** (calcium-dependent phospholipid binding).
- **Cellular processes.** Presynaptic vesicle fusion, endocytic recycling, and — in the recessive form — ultrastructural remodeling of the nerve terminal (F006).
- **Protein dysfunction.** Dominant = dominant-negative interference; recessive = loss of function. C2B Ca^{2+} -binding-pocket modeling (SwissModel) predicts structural disruption for pathogenic variants (F009; [PMID: 33320396](#), [PMID: 28953919](#)).
- **Metabolic / immune involvement.** None — no metabolic derangement and **no autoimmunity** (ACh-receptor and VGCC antibodies negative), which distinguishes CMS7 from MG and autoimmune LEMS (F011).
- **Molecular profiling (electrophysiology & ultrastructure).** Microelectrode studies in the dominant form show **markedly reduced quantal content that increases linearly with higher stimulation frequencies**; MEPP frequency is normal at rest but rises with

stimulation. Electron microscopy shows **overdeveloped postsynaptic folding and abundant endosomes, multivesicular bodies, and degenerative lamellar bodies within small nerve terminals**; presynaptic failure is more prominent in the recessive form (F006; PMID: 34037996).

Upstream vs downstream. Upstream = SYT2 variant → defective Ca^{2+} -sensing/SNARE coupling. Downstream = reduced quantal ACh release → EPP failure → fatigable weakness, with facilitation as a distinctive downstream compensatory readout, and secondary distal neuropathy/terminal remodeling as a chronic downstream consequence.

Cell types & CL terms. Presynaptic **motor neuron / lower motor neuron** terminal (CL:0000100 motor neuron; CL:0011001 spinal cord motor neuron) and the **skeletal muscle fiber** (CL:0000188) postsynaptic partner.

7. Anatomical Structures Affected

- **Organ / system level.** Primary target is the **peripheral nervous system–muscle interface: the neuromuscular junction** (UBERON:0002476, neuromuscular junction; GO:0031594 neuromuscular junction as a cellular component). Body systems: **somatic motor / musculoskeletal** and, secondarily, **respiratory** (respiratory muscle involvement in severe recessive disease) and **bulbar** musculature.
- **Tissue / cell level.** **Skeletal (striated) muscle** (UBERON:0001134) and the presynaptic **motor neuron terminal**. Predominant clinical involvement is **distal lower limb** in the dominant form; generalized/axial and bulbar/respiratory in the recessive form.
- **Subcellular level.** **Synaptic vesicle** and **presynaptic active zone** membranes (GO:0008021 synaptic vesicle; GO:0048786 presynaptic active zone). Ultrastructurally, endosomes, multivesicular bodies, and lamellar bodies accumulate in nerve terminals (F006).
- **Localization / lateralization.** Symmetric, **bilateral** involvement; distal-predominant in the dominant form. Specific sites: lower-limb muscles and feet (pes cavus). UBERON: UBERON:0002103 (hindlimb), UBERON:0002387 (pes/foot).

8. Temporal Development

- **Onset. Congenital / neonatal** in the recessive form (hypotonia, weak cry, feeding/respiratory problems from birth); **childhood-onset** distal weakness typical in the dominant form, with some presentations recognized later. Onset pattern is **chronic/insidious** rather than acute (F005, F012).
 - **Progression.** Dominant form: **stable or slowly progressive**, described in the seminal report as a "*nonprogressive motor neuropathy*" and in the biallelic series as "*stable or slowly progressive distal weakness of variable severity*" (F012; [PMID: 25192047](#), [PMID: 32776697](#)). Recessive form: severe from birth with variable respiratory course.
 - **Course pattern & duration. Chronic, lifelong**, with a **fluctuating/fatigable** functional pattern superimposed on a stable structural baseline. No spontaneous remission is described; symptomatic improvement is **treatment-induced** (3,4-DAP, AChE inhibitors).
 - **Critical periods.** The neonatal/infantile period is the window of greatest vulnerability (respiratory failure risk) and the key window for diagnosis and initiation of treatment in the severe recessive form.
-

9. Inheritance and Population

Epidemiology. CMS7/SYT2-CMS is **ultra-rare**. Genetically confirmed CMS overall prevalence is estimated at **6.5 per million overall and 8.5 per million in the pediatric population** in the UK (n=442 cohort; F004; [PMID: 41251564](#)), with an earlier estimate of **9.2 per million** ([PMID: 34736634](#)). The most common CMS subtypes are CHRNE, DOK7, and RAPSN; **SYT2-CMS is one of the rarest presynaptic subtypes, with only ~10–15 families reported worldwide** across dominant and recessive forms (F004). No incidence figures are available specifically for CMS7.

Inheritance genetics (F010, F013).

Feature	Dominant (CMS7A)	Recessive (CMS7B)
Pattern	Autosomal dominant (multigenerational or de novo)	Autosomal recessive
Variant class	C2B missense / in-frame	Biallelic LoF (frameshift/nonsense/splice/deletion)
Mechanism	Dominant-negative	Loss of function
Penetrance	High (segregates in families; de novo cases)	High
Consanguinity	Not typically	Frequently present
Severity	Milder, slowly progressive	Severe, congenital

De novo dominant occurrence is documented (*"a new de novo heterozygous in frame deletion of the SYT2 gene"*; PMID: 33320396). Recessive disease is documented across multiple consanguineous families (*"we report seven patients of five families, with biallelic loss of function variants in SYT2"*; PMID: 32776697).

Penetrance, expressivity, anticipation, mosaicism, founder effects. Penetrance appears **high**; expressivity is **variable** (severity gradient by allele type). **No genetic anticipation** (not a repeat-expansion disorder). No germline mosaicism or founder effects reported. **Carrier frequency** is not quantifiable — pathogenic variants are absent/singleton in gnomAD (F013).

Population demographics. No ethnic predilection beyond the association of the recessive form with **consanguineous** populations. No sex bias reported; both sexes affected (autosomal). Geographic distribution: cases reported across Europe, the Middle East, and Asia — sporadic, without endemic clustering.

10. Diagnostics

Diagnosis integrates **presynaptic electrophysiology** with **molecular genetics** (F011).

Electrophysiology (the diagnostic hallmark). Repetitive nerve stimulation (RNS) shows **low baseline CMAP amplitude**, a **decremental response at low frequency**, and a **marked increment/facilitation after brief exercise or high-frequency stimulation**, with **prolonged post-tetanic potentiation lasting up to ~60 minutes** and single-fiber EMG jitter/blocking — a

LEMS-like presynaptic signature (F006, F011; [PMID: 26519543](#): "Electrophysiologic testing revealed features indicative of a presynaptic deficit in neurotransmitter release with posttetanic potentiation lasting up to 60 minutes"; [PMID: 29696584](#): "They have low compound muscular action potential amplitude that increment after brief exercise (facilitation) or high-frequency repetitive nerve stimulation").

Laboratory. CK normal or mildly elevated (e.g., 501 U/L; [PMID: 41331967](#)). **ACh-receptor and VGCC antibodies negative** — critical for excluding autoimmune MG/LEMS.

Genetic testing. SYT2 is identified via **whole-exome sequencing** and **multigene CMS/neuromuscular NGS panels**, with Sanger confirmation; **chromosomal microarray/exome** can detect large exonic deletions (the exons 2–9 deletion; [PMID: 36722210](#)). Because CMS is now attributable to ~40 genes, panel/exome testing is the standard first-line molecular approach ([PMID: 40533459](#), [PMID: 36835142](#)). Single-gene SYT2 testing is appropriate when the LEMS-like presynaptic phenotype is recognized. WGS is useful for deep-intronic/structural variants when panel/exome is negative. Mitochondrial DNA testing, karyotyping, FISH, and repeat-expansion testing are **not indicated**.

Muscle biopsy / pathology. Typically shows only **mild/nonspecific neurogenic** features; EM (research setting) shows the terminal remodeling described above (F006; [PMID: 33320396](#)).

Differential diagnosis. Autoimmune LEMS (antibody-positive, often paraneoplastic); other presynaptic CMS (AGRN, VAMP1, UNC13A/MUNC13-1, LAMA5, SLC5A7, SLC18A3); distal hereditary motor neuropathy / Charcot–Marie–Tooth; spinal muscular atrophy; and congenital myopathies (F011; [PMID: 29696584](#), [PMID: 33320396](#)).

Screening. No population newborn screening exists for CMS7. **Cascade genetic testing** of relatives and, where a familial variant is known, prenatal/preimplantation options apply.

11. Outcome / Prognosis

- **Survival / mortality.** No formal survival statistics exist. The dominant form is compatible with normal life expectancy. In the severe recessive form, **respiratory insufficiency and recurrent respiratory infections** are the principal drivers of morbidity and potential mortality (F012; [PMID: 41331967](#), [PMID: 32776697](#)).
- **Morbidity / function.** Dominant form: mobility limitation from distal weakness and foot deformity, fatigue. Recessive form: major disability with feeding/respiratory support needs in infancy.

- **Disease course & recovery.** Symptoms are chronic; **substantial functional improvement is achievable with treatment** even in severe recessive disease — *"Treatment with an acetylcholinesterase inhibitor pursued in three patients showed clinical improvement with increased strength and function"* (PMID: 32776697).
 - **Prognostic factors.** The strongest prognostic determinant is **allele class / inheritance mode** (dominant missense = milder; biallelic LoF = severe), followed by degree of respiratory involvement and treatment responsiveness. No molecular prognostic biomarkers beyond genotype.
-

12. Treatment

CMS7 is **treatable**, and — as in CMS generally — the choice of agent is **subtype-specific** (F003; PMID: 30032336). There is **no role for immunotherapy** (it is not autoimmune).

Therapy	Class / mechanism	Evidence in SYT2-CMS	Suggested NCIT
3,4-Diaminopyridine (amifampridine)	K ⁺ -channel blocker; prolongs presynaptic depolarization → more Ca ²⁺ influx → more ACh release	Clinical benefit + improved neuromuscular transmission in dominant form (PMID: 26519543); ameliorated fatigue in recessive form (PMID: 32250532)	NCIT:C61693 (Amifampridine)
Pyridostigmine (AChE inhibitor)	Prolongs ACh at the endplate	Ameliorated fatigue in recessive case (PMID: 32250532); AChE-inhibitor improved strength/function in biallelic patients (PMID: 32776697)	NCIT:C739 (Pyridostigmine)
Salbutamol / albuterol (β ₂ -agonist)	β-adrenergic modulation of endplate	Mixed/limited: improved patient-reported fatigue but no clear motor/neurophysiologic benefit in NMJ-involving motor neuropathies incl. SYT2 (PMID: 36869887); ineffective in one recessive case (PMID: 32250532)	NCIT:C29082 (Albuterol)

Mechanistically, 3,4-DAP is the rational first-line agent for a presynaptic release deficit because it augments Ca²⁺ influx and quantal release; AChE inhibitors are a useful adjunct/alternative. The pharmacologic logic mirrors treatment of LEMS. Broader CMS treatment principles: *"Cholinergic agents, β-adrenergic agonists, and open-channel blockers remain the principal treatment modalities,"* and pyridostigmine should be **avoided** in DOK7, AChE deficiency, and slow-channel CMS — underscoring the necessity of an accurate genetic diagnosis before treatment (PMID: [30032336](#)).

Supportive / rehabilitative. Respiratory support and infection management (severe recessive disease), physical/occupational therapy, orthotic management of foot deformities, and nutritional/feeding support in infancy.

Advanced/experimental therapeutics. None approved specifically for CMS7. No gene, cell, or RNA therapy exists. A research-stage delivery concept exploits SYT2's luminal domain: intravenously administered **anti-SYT2 antibodies selectively localize to NMJs** and are retrogradely transported to motor neurons — a potential future neuron-targeted delivery route rather than a CMS7 therapy ([PMID: 40454418](#)).

Pharmacogenomics / personalized medicine. Treatment is effectively **genotype-guided** at the level of CMS subtype (presynaptic → 3,4-DAP/AChE inhibitor). No SYT2-specific pharmacogenomic variants are described.

13. Prevention

CMS7 is a Mendelian disorder, so prevention is **genetic**, not behavioral or environmental.

- **Primary prevention.** Not applicable in the public-health sense. The relevant tools are **genetic counseling, carrier testing** of at-risk relatives (recessive form), and reproductive options — **prenatal diagnosis** and **preimplantation genetic testing** — where a familial pathogenic variant is known.
- **Secondary prevention.** Early recognition of the LEMS-like presynaptic electrophysiology → prompt genetic diagnosis → early initiation of 3,4-DAP/AChE inhibitors to prevent avoidable disability; vigilant respiratory monitoring and prompt treatment of respiratory infections in the severe recessive form.
- **Tertiary prevention.** Respiratory support, immunization against respiratory pathogens (influenza, pneumococcus) to reduce infection-triggered crises, orthopedic/rehabilitative management of contractures and foot deformities.
- **Counseling.** Recurrence risk is **50%** for offspring of a dominant-variant carrier and **25%** for siblings in recessive families; consanguinity counseling is relevant.

14. Other Species / Natural Disease

- **Taxonomy / orthologs.** SYT2 is evolutionarily conserved. Mouse **Syt2** (NCBI Gene 20680) is the ortholog and is the major fast-release Ca^{2+} sensor in caudal brain/brainstem and motor neurons (F009; [PMID: 17521570](#), [PMID: 20729199](#)). *Drosophila melanogaster* synaptotagmin

is functionally orthologous for modeling purposes. NCBI Taxa: *Mus musculus* (10090), *Drosophila melanogaster* (7227).

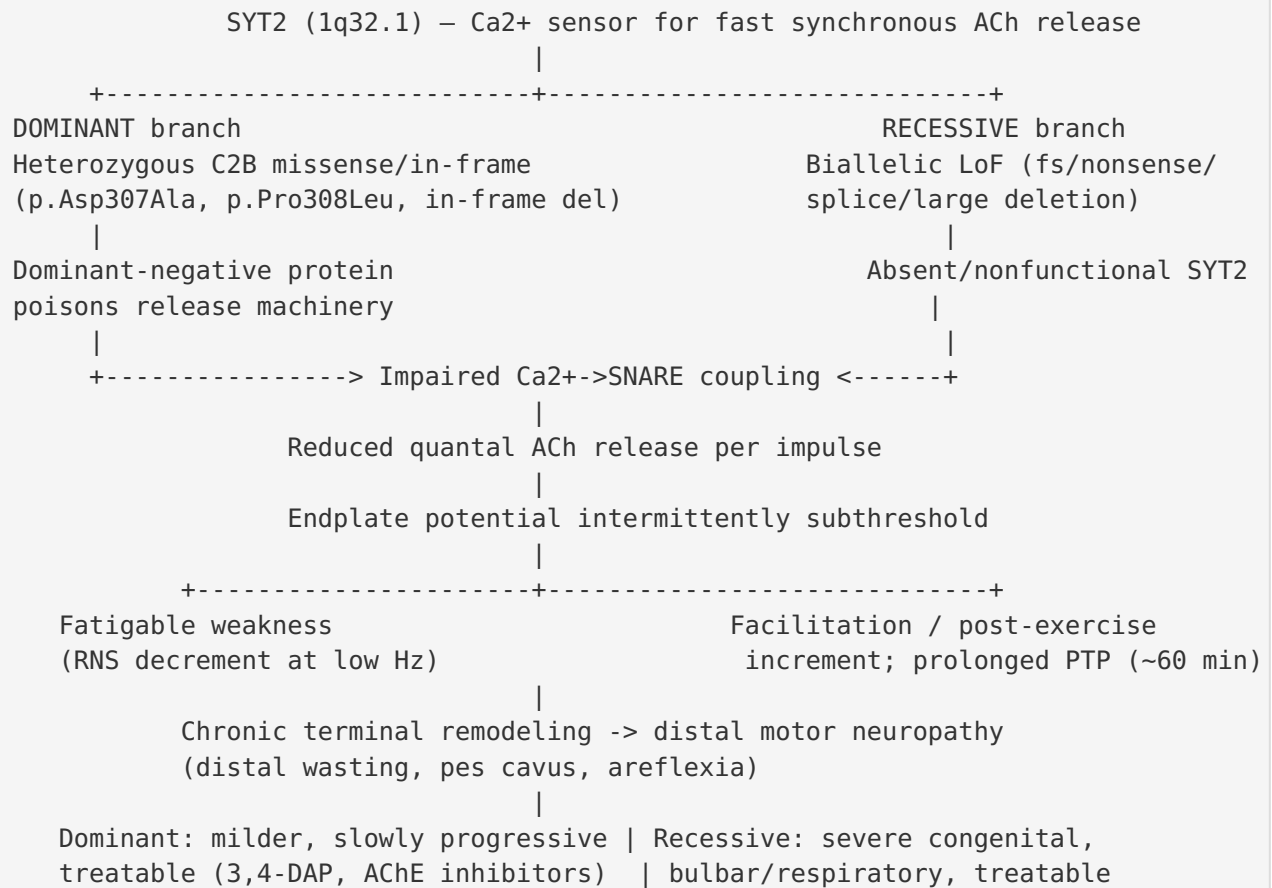
- **Natural disease in other species.** No naturally occurring SYT2 congenital myasthenic syndrome has been reported in companion animals or wildlife (no OMIA entry noted); CMS7 is, to date, a human-described disease with experimental animal models rather than a recognized spontaneous veterinary disease. (Not available.)
- **Comparative biology.** The evolutionary conservation of synaptotagmin's Ca^{2+} -sensor function underpins the validity of invertebrate and rodent models; the fly heterozygous model recapitulates the human dominant-negative phenotype (see Section 15).

15. Model Organisms

- ***Drosophila melanogaster* (primary causality model).** Expressing the human-homologous C2B mutation (P308-equivalent) in flies **lacking native synaptotagmin was lethal**, demonstrating the residue is critical for synaptotagmin function; **co-expression with one wild-type copy** (mimicking the heterozygous human state) produced **neurological/behavioral manifestations similar to patients** and **dominant disruption of synaptic-vesicle exocytosis** — directly establishing pathogenicity and the dominant-negative mechanism (F009; [PMID: 28953919](#): "*When expressed in the absence of native synaptotagmin, this mutation is lethal, demonstrating for the first time that this residue plays a critical role in synaptotagmin function*" and "*Drosophila carrying this mutation developed neurological and behavioral manifestations similar to those of human patients*").
- **Mouse (Sytn2).** The mouse ortholog is the validated fast-release Ca^{2+} sensor for brainstem/motor-neuron synapses ([PMID: 17521570](#)); Sytn2-manipulated rodents inform normal SYT2 physiology, though a dedicated CMS7 patient-variant knock-in mouse is not prominently reported.
- **In silico / structural modeling.** C2B Ca^{2+} -binding-pocket modeling (SwissModel; rat Sytn1 C2B X-ray template) predicts structural disruption for pathogenic variants and supports variant interpretation ([PMID: 33320396](#), [PMID: 32250532](#)).
- **Phenotype recapitulation & limitations.** The fly model faithfully reproduces the **dominant-negative exocytosis defect and dominant inheritance**; it does not capture the human distal motor-neuropathy/foot-deformity phenotype or the recessive LoF disease, and mammalian models specifically engineered with patient recessive alleles are lacking.

Applications: dissecting Ca^{2+} -sensing/SNARE coupling, dominant-negative mechanism, and candidate therapeutics.

Mechanistic Model / Interpretation



The unifying concept is that **CMS7 is a disorder of presynaptic quantal release** in which allele class dictates severity: a poisoning (dominant-negative) mechanism produces a milder, distal, neuropathy-mimicking disease, while complete loss of the fast Ca^{2+} sensor produces severe congenital myasthenia. The **facilitation** phenomenon — pathognomonic on electrophysiology — is the direct downstream readout of residual Ca^{2+} rescuing release during repetitive activity, and it is precisely this mechanism that **3,4-DAP exploits therapeutically**.

Evidence Base

PMID	Role in this report
25192047	Seminal disease-gene report; dominant C2B missense variants; AD LEMS-like + nonprogressive motor neuropathy
26519543	Electrophysiology (presynaptic deficit, prolonged PTP) and 3,4-DAP treatability in dominant form
34037996	Dominant vs recessive comparison; quantal-content and EM ultrastructural findings
32776697	Biallelic LoF recessive CMS7; AChE-inhibitor benefit; stable/slowly progressive course
32250532	Recessive p.Arg397Serfs*37; 3,4-DAP/pyridostigmine effective, albuterol ineffective; C-terminal SNARE-interaction modeling
33659639	New homozygous recessive variants (c.465+1G>A, c.328_331dup); myopathy-mimicking presentation
33320396	De novo dominant in-frame deletion; CMT/neuropathy mimic; genetic-testing recommendation
36722210	Large exonic deletion (exons 2–9) detected by exome+microarray; consanguineous recessive
28953919	Drosophila model establishing causality and dominant-negative mechanism
17521570	SYT1/2/9 as fast-release Ca ²⁺ sensors; SYT2 as major NMJ isoform
29696584	Places SYT2 in the CMS-LEMS presynaptic group; defines facilitation hallmark
30032336	CMS treatment principles; subtype-specific drug choice; no immunotherapy
41331967	28-case review; core phenotype; C2B clustering; representative case features
41251564	UK CMS prevalence (6.5/8.5 per million)
34736634	Earlier CMS prevalence (9.2 per million); >30 genes
40533459 / 36835142	40- and 35-gene CMS reviews; panel/exome testing standard
36869887	Salbutamol trial in NMJ-involving motor neuropathies incl. SYT2
20729199	SYT2 regulation/expression biology (calmodulin, rostral–caudal)

PMID	Role in this report
40454418	Anti-SYT2 luminal-domain antibody NMJ targeting (future delivery concept)

All quoted snippets above are verbatim from the corresponding abstracts as recorded in the knowledge state (findings F001–F013).

Limitations and Knowledge Gaps

1. **Extreme rarity → weak epidemiology.** Only ~10–15 families are reported worldwide; there are **no CMS7-specific prevalence/incidence figures, no natural-history registries, and no formal survival/QoL data.** Prevalence is inferred from all-CMS studies.
2. **No controlled treatment trials.** Efficacy of 3,4-DAP and AChE inhibitors rests on **case reports and small series**, not randomized data. Salbutamol data are mixed and derive from a heterogeneous cohort.
3. **Genotype–phenotype correlation incomplete.** The dominant-negative vs LoF dichotomy is well supported, but finer correlations (specific variant → severity/organ involvement) remain anecdotal.
4. **Model-organism gaps.** Causality is proven in *Drosophila*; a mammalian knock-in carrying a patient dominant allele, and models of the recessive LoF disease, are lacking — limiting preclinical therapeutic testing.
5. **No modifier/epigenetic data.** No modifier genes, epigenetic marks, or environmental modifiers have been identified.
6. **QoL/functional instruments unmeasured.** No EQ-5D/SF-36/PROMIS or ICF-based disability data specific to CMS7.

Proposed Follow-up Experiments / Actions

1. **International SYT2-CMS registry.** Aggregate all reported and unpublished cases to derive genotype-stratified natural history, treatment-response rates, and survival — the single highest-value action given rarity.

2. **Prospective, protocolized 3,4-DAP ± pyridostigmine response study.** Standardized RNS/CMAP endpoints plus patient-reported fatigue to formalize the treatment evidence base (currently case-level).
3. **Patient-variant knock-in mouse models.** Generate C2B dominant-negative (e.g., p.Asp307Ala) and biallelic-LoF mice to reproduce distal neuropathy and severe congenital disease respectively, and to test therapeutics.
4. **Structure-function dissection of C2B variants.** Combine cryo-EM/AlphaFold modeling with electrophysiology in reconstituted or iPSC-derived motor-neuron/NMJ systems to map how each variant perturbs Ca^{2+} binding and SNARE coupling.
5. **iPSC-derived motor neuron / NMJ-on-chip modeling.** Patient-derived systems to screen candidate presynaptic-release enhancers beyond 3,4-DAP.
6. **Explore SYT2-targeted delivery.** Evaluate the anti-SYT2 luminal-domain NMJ-targeting antibody shuttle ([PMID: 40454418](#)) as a route for future gene- or protein-replacement approaches to the recessive LoF form.
7. **Refine diagnostic pathways.** Promote recognition of the LEMS-like presynaptic RNS signature in antibody-negative patients to shorten time-to-genetic-diagnosis and ensure SYT2 is on CMS/neuropathy NGS panels.

Report compiled from 13 confirmed findings (F001–F013) across 29 reviewed publications. Evidence types: predominantly human clinical case reports/series and comprehensive reviews, supplemented by a Drosophila causality model, rodent/in-vitro SYT2 physiology, and computational (gnomAD constraint, structural modeling) analyses.
